PRAKTIKUM BASIS DATA

MODUL 8

Scripting SQL Create

JURNAL

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1. Desain Database

Tabel Department:

```
--MEMBUAT TABEL
CREATE TABLE DEPARTMENTS (
    department_id NUMBER PRIMARY KEY,
    department_name VARCHAR2(100),
    manager_id NUMBER,
    location_id NUMBER
);
```

Tabel Employee:

```
CREATE TABLE EMPLOYEES (
    employee_id NUMBER PRIMARY KEY,
    last_name VARCHAR2(100) NOT NULL,
    email VARCHAR2(100),
    salary NUMBER,
    commission_pct NUMBER,
    hire_date DATE NOT NULL,
    department_id NUMBER,
    CONSTRAINT fk_dept
        FOREIGN KEY (department_id)
        REFERENCES DEPARTMENTS(department_id)
);
```

Tabel tambahan:

```sql
CREATE TABLE SUPPLIERS (
    supplier_id NUMBER PRIMARY KEY,
    supplier_name VARCHAR2(100),
    contact_email VARCHAR2(100)
);

CREATE TABLE PRODUCTS (
    product_id NUMBER PRIMARY KEY,
    product_name VARCHAR2(100),
    supplier_id NUMBER,
    price NUMBER,
    stock NUMBER,
    CONSTRAINT fk_supplier_prod
        FOREIGN KEY (supplier_id)
        REFERENCES SUPPLIERS(supplier_id)
);
```

2. Analisis tabel dengan compute statistic:

```sql
--ANALISIS STATISTIK TABEL
ANALYZE TABLE DEPARTMENTS COMPUTE STATISTICS;
ANALYZE TABLE EMPLOYEES COMPUTE STATISTICS;
ANALYZE TABLE SUPPLIERS COMPUTE STATISTICS;
ANALYZE TABLE PRODUCTS COMPUTE STATISTICS;
```

3. Insert data:

Insert department:

```sql
--INSERT DATA
INSERT INTO DEPARTMENTS VALUES (10, 'HR', 101, 1000);
INSERT INTO DEPARTMENTS VALUES (20, 'IT', 102, 1001);
INSERT INTO DEPARTMENTS VALUES (30, 'Finance', 103, 1002);
INSERT INTO DEPARTMENTS VALUES (40, 'Marketing', 104, 1003);
INSERT INTO DEPARTMENTS VALUES (50, 'Sales', 105, 1004);
INSERT INTO DEPARTMENTS VALUES (60, 'Logistics', 106, 1005);
INSERT INTO DEPARTMENTS VALUES (70, 'Legal', 107, 1006);
INSERT INTO DEPARTMENTS VALUES (80, 'Operations', 108, 1007);
INSERT INTO DEPARTMENTS VALUES (90, 'Support', 109, 1008);
INSERT INTO DEPARTMENTS VALUES (100, 'Admin', 110, 1009);
```

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	HR	101	1000
2	20	IT	102	1001
3	30	Finance	103	1002
4	40	Marketing	104	1003
5	50	Sales	105	1004
6	60	Logistics	106	1005
7	70	Legal	107	1006
8	80	Operations	108	1007
9	90	Support	109	1008
10	100	Admin	110	1009

Insert Employee:

```
INSERT INTO EMPLOYEES VALUES (1, 'Corey', 'coreytaylor@mail.com', 5000, 0.1, SYSDATE, 80);
INSERT INTO EMPLOYEES VALUES (2, 'James', 'jamesroot@mail.com', 6000, 0.2, SYSDATE, 80);
INSERT INTO EMPLOYEES VALUES (3, 'Mick', 'mickthomson@mail.com', 5500, NULL, SYSDATE, 20);
INSERT INTO EMPLOYEES VALUES (4, 'Paul', 'paulgray@mail.com', 7000, 0.15, SYSDATE, 30);
INSERT INTO EMPLOYEES VALUES (5, 'Joey', 'joeyjordison@mail.com', 4500, NULL, SYSDATE, 10);
INSERT INTO EMPLOYEES VALUES (6, 'Shawn', 'shawncrahan@mail.com', 4800, 0.05, SYSDATE, 80);
INSERT INTO EMPLOYEES VALUES (7, 'Chris', 'chrisfehn@mail.com', 5200, NULL, SYSDATE, 50);
INSERT INTO EMPLOYEES VALUES (8, 'Sid', 'sidwilson@mail.com', 6100, 0.1, SYSDATE, 80);
INSERT INTO EMPLOYEES VALUES (9, 'Craig', 'craigjones@mail.com', 5300, NULL, SYSDATE, 60);
INSERT INTO EMPLOYEES VALUES (10, 'Slip', 'slipknot@mail.com', 4900, 0.05, SYSDATE, 80);
```

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTMENT_ID
1	1	Corey	coreytaylor@mail.com	5000	0,1	15-04-2026	80
2	2	James	jamesroot@mail.com	6000	0,2	15-04-2026	80
3	3	Mick	mickthomson@mail.com	5500	(null)	15-04-2026	20
4	4	Paul	paulgray@mail.com	7000	0,15	15-04-2026	30
5	5	Joey	joeyjordison@mail.com	4500	(null)	15-04-2026	10
6	6	Shawn	shawncrahan@mail.com	4800	0,05	15-04-2026	80
7	7	Chris	chrisfehn@mail.com	5200	(null)	15-04-2026	50
8	8	Sid	sidwilson@mail.com	6100	0,1	15-04-2026	80
9	9	Craig	craigjones@mail.com	5300	(null)	15-04-2026	60
10	10	Slip	slipknot@mail.com	4900	0,05	15-04-2026	80

4. Analisis tabel dengan compute statistic:

```
--ANALISIS STATISTIK LAGI
ANALYZE TABLE DEPARTMENTS COMPUTE STATISTICS;
ANALYZE TABLE EMPLOYEES COMPUTE STATISTICS;
ANALYZE TABLE SUPPLIERS COMPUTE STATISTICS;
ANALYZE TABLE PRODUCTS COMPUTE STATISTICS;
```

5. Buat view empvu80:

```
--MEMBUAT VIEW EMPVU80
CREATE VIEW EMPVU80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM EMPLOYEES
WHERE department_id = 80;
```

6. Tampilkan struktur view:

```
--MENAMPILKAN STRUKTUR VIEW
SELECT column_name, data_type
FROM user_tab_columns
WHERE table_name = 'EMPVU80';
```

	COLUMN_NAME	DATA_TYPE
1	ID_NUMBER	NUMBER
2	NAME	VARCHAR2
3	SALARY	NUMBER
4	DEPARTMENT_ID	NUMBER

7. Buat non-unique index:

```
--MEMBUAT NOT UNIQUE INDEX
CREATE INDEX idx_emp_dept
ON EMPLOYEES(department_id);
```